Fiscal Year:	FY 2015	**Task Last Updated:**	FY 12/17/2015
PI Name:	Hollberg, Leo W Ph.D.		
Project Title:	Advancing Time Transfer and Optical Atomic Clocks for Space		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Stanford	**State:**	CA
Zip Code:	94305-4008	**Congressional District:**	15
Comments:	NOTE: Previously at National Institute of Standards & Technology (Ed., Fall 2011)		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	05/30/2012	**End Date:**	05/30/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1458931		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Our research project focuses on one of the most critical and significant scientific and technical challenges to using advanced cold-atom atomic clocks in space. The problems that we address are the serious performance limitations to the existing Time-Transfer links between ground and space. This research will advance both the basic science and the technology of Time-Transfer methods that could both support the current European Space Agency-Atomic Clock Ensemble in Space (ESA-ACES) mission plan and subsequently enhance performance in order to meet the demands of future optical atomic clocks (e.g., ESAs-SOC) for space. The improved performance is enabled by advanced laser methods for absolute Time and Frequency transfer that are based on stabilized femtosecond mode-locked lasers. We have developed a novel integrated system concept that will optimize the Time-Transfer performance in real-world systems by combining an ultrahigh performance Laser Time-Transfer (Laser-TT) link with an advanced microwave link and GPS signals. Analysis of the fundamental capabilities and limitations will guide the final system design, and a		

	prototype Laser-TT link will be built in the laboratory and tested to critically evaluate the system performance. The demonstrations and testing of the Time-Transfer link will include two-way laser transmission via both free-space and optical fibers, and two-way microwave transmission via coaxial cable, which will give over-constrained redundant signals allowing us to identify systematic errors and uncertainties. Our Laser-TT link will improve the absolute timing precision by a factor of 100 times compared to the most advanced microwave links. Nonetheless, the highest performance, practical, and reliable systems operating in real-world conditions (with weather, turbulence, etc.) will require both microwave and optical channels. Critical analysis and testing of the most promising concepts will allow us to optimally integrate an advanced microwave link (as per ACES) with our higher performance Laser-TT link.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced precision for "Time" (epoch), time interval, and frequency transfer to space could be enabling for several science and technology missions and applications, including: International Space Station (ISS), ACES, tests of relativity, geodesy, gravity. Optical timing can improve the performance and precision of Global Satellite Navigation Systems (GNSS) such as GPS. It could also provide much improved precision and accuracy of precise orbit determination and through better timing and orbit could enhance capabilities and accuracy in geodesy. Potentially leverage capabilities with existing International Laser Ranging Systems to mutual benefit. Definitely strong synergism with satellite laser communication systems now under development.
Task Progress:	This project focuses on developing laser methods and related technologies to transfer precise Time and frequency signals through free-space optical links between ground and space. Highly accurate timing is necessary for space tests of Einstein's General Relativity [e.g.: measurements of the gravitational Red Shift, tests of Local Lorentz Invariance (LLI), tests of Local Position Invariance (LPI), potential for Gravitational wave detection, and search for physics beyond the Standard Model (SM)]. Our goals are to provide precise timing worldwide at the 1 ps level and ranging between ground and space with 1 mm accuracy. Our program is strongly motivated by proposed future missions that would put high performance atomic clocks in space. In part, this project is motivated by the upcoming ACES mission of ESA, and perhaps our approach will provide higher performance time transfer than alternative practical methods. Our approach would leave the best atomic clocks on the ground at national standards labs (e.g. NIST, LNE-SYRTE, PTB) and use fs optical frequency combs (FOFC) referenced to those atomic frequency standards to encode and transfer the precisely timed optical signals with a two-way lasercom optical communication link. During this phase of our effort we have developed viable and practical system architecture that can meet our timing and ranging goals. This system approach will also enable new capabilities and enhance performance in geodesy and Earth sciences and perhaps existing GNSS systems. During this third year of funding from NASA Fundamental Physics, our research on Time-transfer and Space-Time references has been presented at two university physics colloquia (University of Nevada, Reno, and Colorado School of Mines) and at three conferences.
Bibliography Type:	Description: (Last Updated: 09/03/2025)
Abstracts for Journals and Proceedings	Berceau P, Hollberg L. "Concept for a space-time reference via laser-comm link." 2015 Joint Conference of the International Frequency Control Symposium and the European Frequency and Time and Forum, Denver, Colo., April 12-16, 2015. Abstracts of 2015 IEEE FCS and EFTF conferences, Denver, Colo., April 12-16, 2015. , Apr-2015
Abstracts for Journals and Proceedings	Berceau P, Hollberg L. "Precise time and frequency transfer using mode-locked femtosecond laser." Annual Symposium of the Stanford Center of Position Navigation and Time. Stanford's 2014 PNT Challenges and Opportunities Symposium, Stanford University, October 2014. Stanford's 2014 PNT Challenges and Opportunities Symposium, Stanford University, October 2014. http://scpnt.stanford.edu/pnt/pnt2014.html#student_schedule , Oct-2014
Articles in Other Journals or Periodicals	Berceau P, Taylor M, Kahn JM, Hollberg L. "Space-time reference with an optical link." Posted Dec 2015 and submitted for publication on arXiv; http://arxiv.org/abs/1512.01912 , Dec-2015